Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 6
Subject: TCD: My Maker Mark	**Date: Week 11 (9 Nov to 13 Nov)**
Class Description: Grade 6 TCD class, My Maker Mark Session 3 of 3 (2 hours). Students finish their laser-cut maker mark using the Unimat Cool Tools sanding machine and drill, then check and export their Junior Engineer product from Week 4 so it can be 3D printed, completing both projects.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Unimat Cool Tools and 3D Print	
Standard(s)	Science: properties of materials; Technology: safe use of machine tools, CAD to 3D print workflow; Math: measurement and precision	
Learning Objective(s) (WALT): We are learning to use small machine tools safely to finish a product to a high quality, and to prepare a model for 3D printing.		**Success Criteria (WILF):** I can name the Unimat Cool Tools machines and what each one does. I can sand and finish my maker mark safely using the sanding machine. I can follow the safety rules for machine tools every time. I can check and export my Junior Engineer model for printing.
Literacy Strategies	Machine tool picture cards; safety pledge read aloud; frame: I used the … to … safely by …	
Assessment	Finished maker mark checked against the quality checklist; tool safety observation; exported STL; live quiz on the TCD website (counts towards the My Maker Mark certificate).	
Targeted Vocabulary	Unimat, sanding machine, jigsaw, lathe, drill, finish, grit, STL, quality	

Skills	Safe tool use, fine motor control, quality control, perseverance
National Identity Integration	Link to Emirati craftspeople such as dhow builders, who shape and finish wood by hand with great care, and to the UAE's growing advanced manufacturing sector.
AI/Digital Learning	Tinkercad; slicer on the teacher's computer; hook video; live quiz on the TCD website. Presentation: Grade6_Week11_Presentation.pptx. Worksheet: Grade6_Week11_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (15 minutes)	Warm-up: Rough or Smooth? (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (25 minutes)	Introduce the Unimat Cool Tools (slide 6). Teach machine tool safety; students read the safety pledge aloud (slides 7 and 11). Demonstrate sanding and finishing (slide 8).
We Do (15 minutes)	Students practise on an offcut at the sanding machine one at a time while the teacher watches. Class reviews the print-ready checklist for the Junior Engineer model (slide 9).
You Do (50 minutes)	Rotation: half the class finishes maker marks at the machines while the other half checks and exports their Junior Engineer models; then swap (slides 12 and 10). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Sand by hand with sandpaper first, then use the sanding machine with the teacher beside you.** **On: Finish your piece on the sanding machine and export your model independently.** **Above: Use the hand drill to add a decorative detail, and write a short quality report comparing your piece with a professional product.**
Closure/ Reflection (15 minutes)	Maker mark showcase: each student shows one detail they finished. Exit ticket (slide 15). The teacher prints Junior Engineer models this week.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: picture cards for each machine and the safety pledge. SEN: hand sanding first and adult beside the student at the machine. G&T: decorative drill detail and

quality report.

Inclusion / EAL Support

- ☑ Visual Supports
- ☑ Chunked Instructions
- ☐ Reduced Language Load
- ☑ Model / Worked Example
- ☐ Pre-teach Vocabulary
- ☑ TA/LSA Support
- ☐ Extra Processing Time
- ☐ Text Read by TA/LSA
- ☐ Flexible Grouping
- ☐ Assistive Technology
- ☐ Movement Break
- ☐ Scaffold / Sentence Starters
- ☐ Other:

G&T / Extension

- ☑ Extension / Challenge Task
- ☐ Higher-order Questioning
- ☑ Depth & Complexity Task
- ☐ Independent Inquiry
- ☐ Leadership / Peer Coaching
- ☑ Real-World Application
- ☐ Curriculum Compacting
- ☐ Above-grade-level Texts
- ☐ Student-led / Passion Project
- ☐ Open-Ended Task
- ☐ Other:

Assessment Adaptations

- ☑ Oral Response
- ☐ Reduced Questions
- ☐ Scaffolded Success Criteria
- ☐ Extra Time
- ☐ Alternative Outcome Format
- ☑ Teacher Conferencing/Live Feedback
- ☐ Quiet / Separate Space
- ☐ Other: